

- 5 D As the object is in static equilibrium, the resultant force of the three external forces acting on the object is equal to zero.

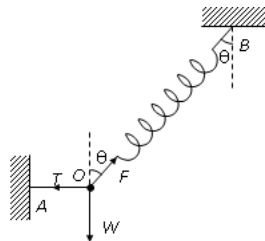
i.e. $\vec{T} + \vec{F} + \vec{W} = 0$

and $\vec{F} + \vec{W} = -\vec{T}$

vertical: $F \cos \theta = W = mg$

horizontal: $F \sin \theta = T$

$\Rightarrow T = mg \tan \theta$



At the instant the thread is suddenly cut, the force $\vec{T}$ is removed. Resultant force acting on the object

$$= \vec{F} + \vec{W}$$

$$= -\vec{T}$$

Hence, acceleration of the object is direction II and magnitude of acceleration = $g \tan \theta$

- 6 B Total momentum of wagon remain unchanged because the vertical force by rain does not affect